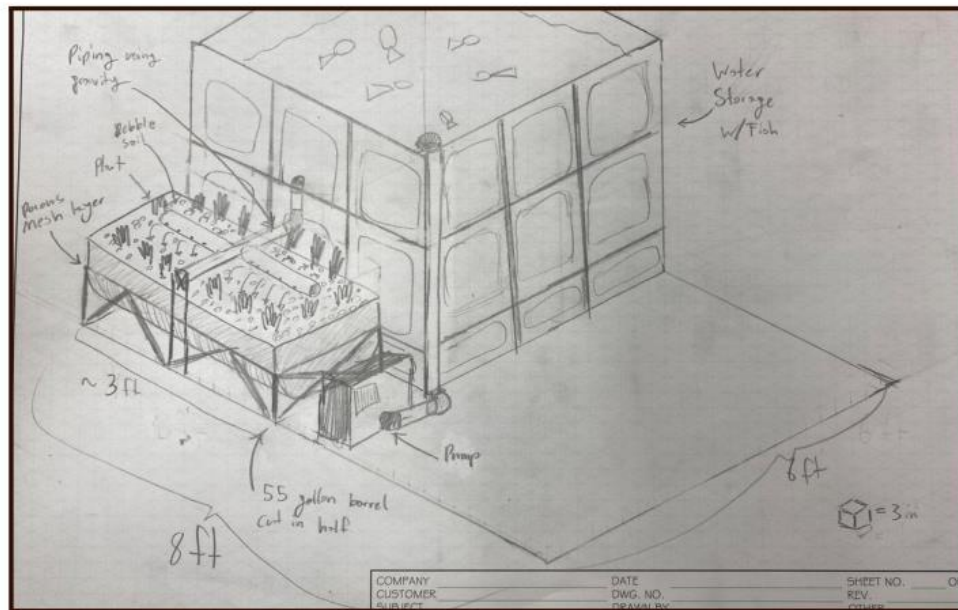
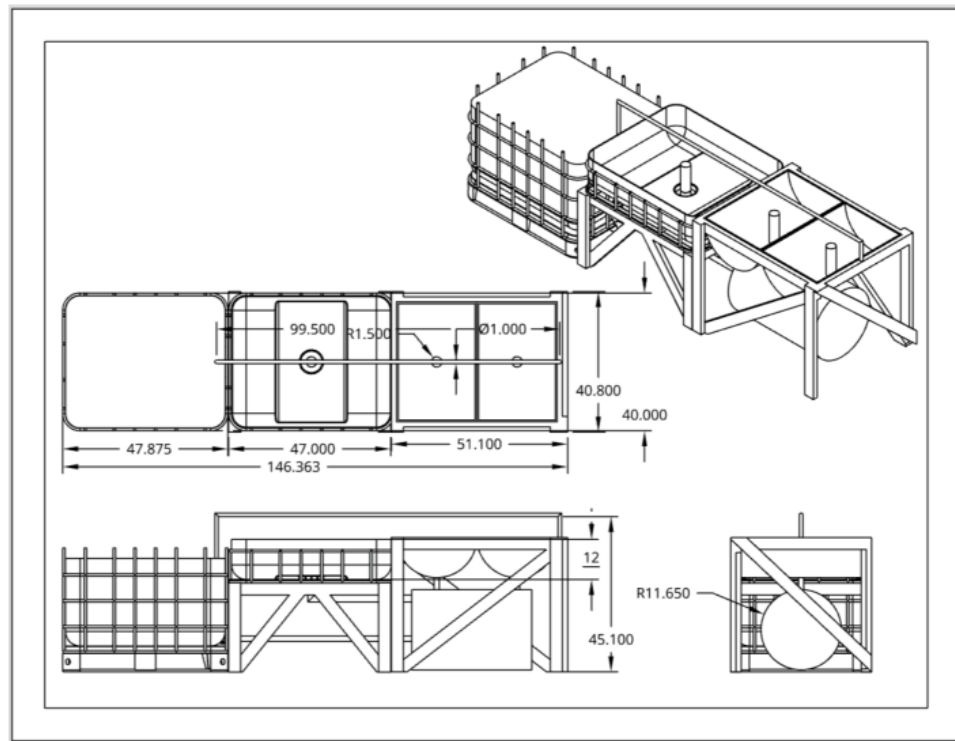


Element G:

Prototype Design



Prototype Iteration:



Explanation of prototype:

We have made some minor adjustments to our design in order to smoothly integrate the aquaponic system indoors, following the Colonial School District's acceptance of this project and their decision on the final location for the system. The system contains a fish tank that is aerated and kept in the back of the structure. The front bed is the 2 halves of the 55 Gallon Barrels that is filled with hydroton rocks that nourish the plants inside. The barrels are also hooked up with air stones to aerate the bed with oxygen. Underneath the grow beds is the slump tank. The slump tank is to store back up water for the smooth efficiency of the system. The main pump is placed in this tank to pump all the water throughout the system, which is all dumped back into the slump tank to be cycled again. The final tank at the front is another type of grow bed which uses styrofoam to float on top of the aired water and deep rooted plants, like lettuce, to then grow freely on top. The whole system is designed to expand and can sustine multiple grow beds

other than the 3 in built in. We have overflow pipes that ensure the system will run hands free without worry of overflowing certain sections of the aquaponic system.

All requirements that can be tested mathematically	In-Depth Analysis	Units Used
Scale and Size	Determining an appropriate scale and size ensures the system is manageable for maintenance and operation within the school setting. It affects factors such as water volume and nutrient cycling efficiency. A good thing to know is the more water volume there is, the easier it is to maintain.	Dimensions (e.g., m ²)
Sustainability	Sustainability is about practices that conserve resources, minimize environmental impact, and promote long term viability. This includes efficient water use and energy conservation.	Measurable by the amount of times that resources needed to be replaced.
Monitoring and Control	Control systems enable students to track key factors such as water quality, temperature, pH, and nutrient levels. This allows for a deeper understanding of the aquaponics system and allows for adjustments to increase the performance of the system.	Various (e.g., pH, °C)

The scale of the system is taller enough to not be of burden on people harvesting or monitoring the system. It is also short enough to not make monitoring and harvesting from the system difficult.

The system uses roughly 500 gallons of water to run for its entire lifespan. It produces roughly 5 LBS of edible biomass per day, once we get off the 1st month start phase. Water can be topped off every once a month to maintain water levels.

Arduino plans are being envisioned for the near future and sensors are being considered to make monitoring easier and quicker. We can monitor the Ph levels and water levels easier all on an app.

Design Requirement	STEM Application	Metrics/Numbers	Expert reviewed comments
Educational Value	Interactive learning modules on aquaponics principles; allows for data collection for student analysis.	Measure the students average grades in science related topics before and after the aquaponics system was implemented.	Mr. Rickert - The students will benefit from the physical system, which helps students who learn that way.
Safety Considerations	Use of non-toxic materials; secure and safe housing for electrical components.	All materials meet ASTM safety standards; electrical components rated for low voltage (12V).	Mr. Rickert- Safety is the most important thing in a school, and the use of non-toxic materials and low voltage electrical components will help with that.
Accessibility	Design accommodating students of various heights and abilities.	Adjustable viewing platforms; braille labels for plants for blind students	Mr. Rickert- There are many different students with a range of physical abilities, and they should all be able to benefit from the system. Adjustable viewing heights will accommodate that.

The system has incredibly educational value, as some students benefit largely from a physical prototype to go along with their learning from the teacher. By providing them with this system, it will improve their understanding of environmental sciences.

Safety considerations are the most important part of a school, as there are many safety implementations all around the building already. We want to make sure that the system we are implementing isn't going to put the teachers or students at risk of any injuries.

We want to make sure that all students are able to benefit from the system, so adding moveable height platforms so that students potentially in wheelchairs will be able to see is crucial.